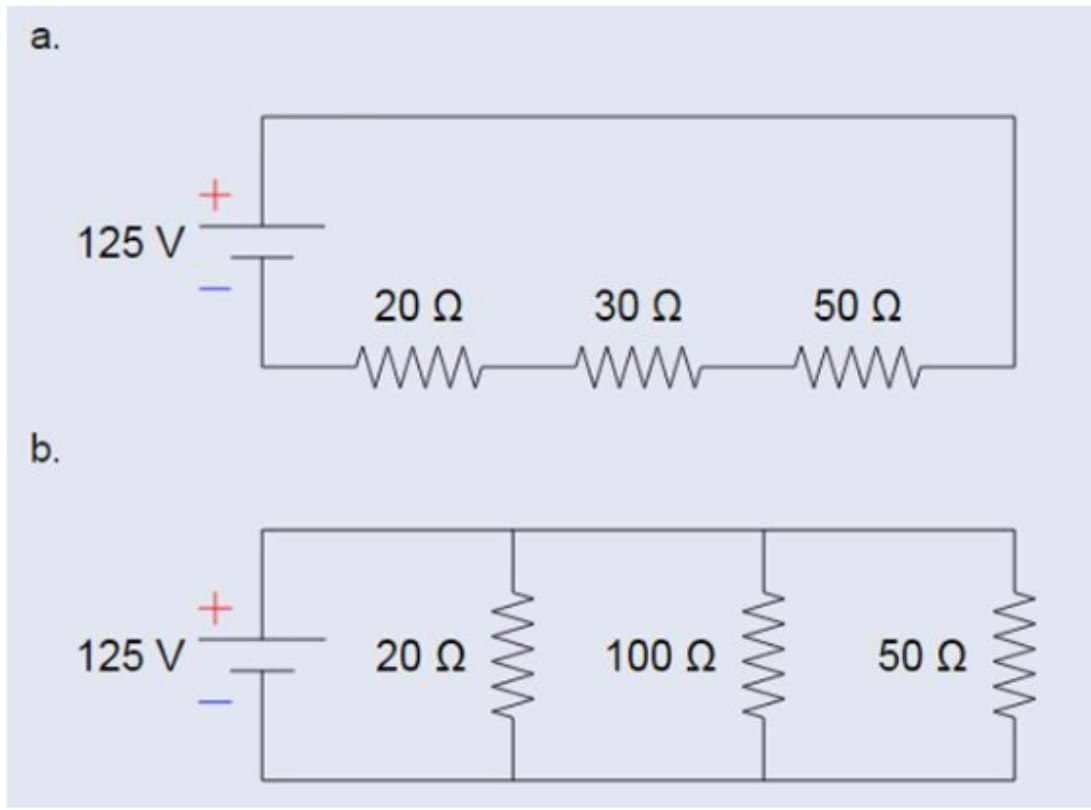


Practical assignment

Part A:



For each of these circuits:

1. Calculate the total resistance in ohms,

$$R(\text{total}) = R_1 + R_2 + R_3$$

$$R(\text{total}) = 20 + 30 + 50 = 100$$

Therefore, total resistance is 100 Ω

2. Calculate the total current from the power supply,

Using Ohm's Law, $I = V/R$

$$I = 125/100 = 1.25$$

Therefore, the total current from the power supply is 1.25 A

3. Calculate the current through each resistor

Because this is a series circuit, the current remains constant throughout the entire path. The current through all three resistors is exactly 1.25A.

4. Calculate the Voltage drop through each resistor

For each resistor, we multiply the current by the resistance

$$20\Omega = 1.25 \times 20 = 25V$$

$$30\Omega = 1.25 \times 30 = 37.5V$$

$$50\Omega = 1.25 \times 50 = 62.5V$$

Also, $25 + 37.5 + 62.5 = 125V$, giving us our original 125 V, therefore confirming this is correct.

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

$$P = V \times I$$

$$20\Omega = 25 \times 1.25 = 31.25W$$

$$30\Omega = 37.5 \times 1.25 = 46.875W$$

$$50\Omega = 62.5 \times 1.25 = 78.125W$$

